

Goals of Security

The **goals of security** define what a secure information system must achieve to protect data and resources from attacks and unauthorized access.

1. Confidentiality

Confidentiality ensures that information is **accessible only to authorized users**. Unauthorized users should not be able to read sensitive data.

Example:

- Encryption of messages
 - Password protection
-

2. Integrity

Integrity ensures that data is **accurate, complete, and not altered** by unauthorized users during storage or transmission.

Example:

- Hash functions
 - Digital signatures
-

3. Availability

Availability ensures that information and services are **available to authorized users whenever required**.

Example:

- Backup systems
 - Protection against Denial of Service (DoS) attacks
-

4. Authentication

Authentication verifies the **identity of a user or system** before granting access.

Example:

- Username and password
 - Biometrics
-

5. Non-Repudiation

Non-repudiation ensures that a sender **cannot deny** sending a message or performing a transaction.

Example:

- Digital signatures
 - Transaction logs
-